

Sky Voice AI — Desktop (**frontend-exe/**)

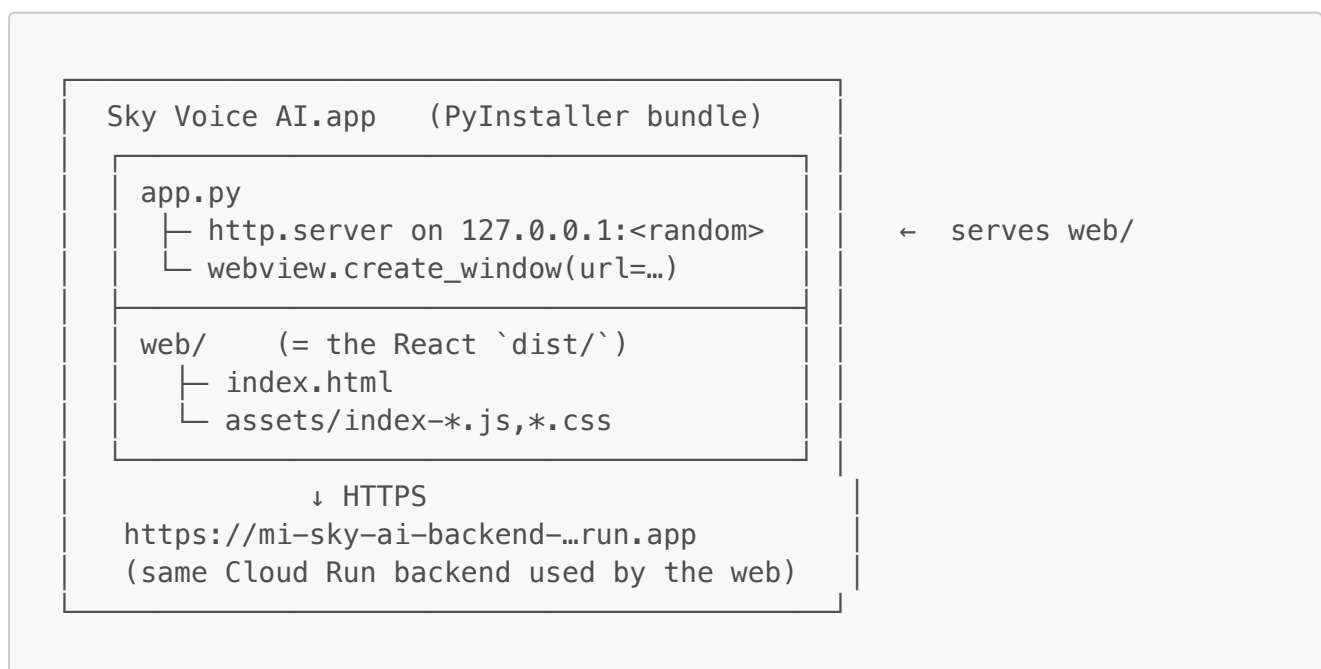
A native, double-clickable build of the operator console — same UI, same code, same Cloud Run backend as the web app at skyvoice.web.app. The React bundle is *reused 1:1* (not rewritten) and embedded inside a [pywebview](#) window, then packaged with PyInstaller.

OS	Output
macOS 11+	dist/Sky Voice AI.app
Windows 10/11	dist/SkyVoiceAI/SkyVoiceAI.exe
Linux (GTK / Qt)	dist/SkyVoiceAI/SkyVoiceAI

Why a wrapper, not a Python rewrite?

The frontend is React + TypeScript + Twilio Voice SDK + WebRTC. Re-implementing it in pure Python would (a) lose the Twilio Voice SDK which only ships for JS, (b) break parity with the web app on every future change, (c) duplicate ~10k lines of UI code. The wrapper approach gives the user a real **.app/.exe**, while every fix shipped to the web app gets picked up by the next desktop build automatically.

Architecture



The local 127.0.0.1 server is required because the Twilio Voice SDK won't grant `getUserMedia` (microphone) on a `file://` origin. Serving

from 127.0.0.1 is treated as a secure context by every modern WebView engine.

Quick start

Run from source (no build, fastest)

```
cd frontend-exe
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt

# Option A – point the window at the live deployed app:
SKYVOICE_URL=https://skyvoice.web.app python3 app.py

# Option B – point it at a local vite dev server (live-reload):
cd ../frontend && npm run dev & # starts http://localhost:5173
cd ../frontend-exe
SKYVOICE_URL=http://localhost:5173 python3 app.py

# Option C – bundled mode (need to build the frontend once):
cd ../frontend && npm run build
cp -R dist ../frontend-exe/web
cd ../frontend-exe && python3 app.py
```

Build the .app / .exe

macOS / Linux:

```
cd frontend-exe
./build.sh # ~2-5 min on first run
open "dist/Sky Voice AI.app" # launch
```

Windows:

```
cd frontend-exe
build.bat
dist\SkyVoiceAI\SkyVoiceAI.exe
```

build.sh / build.bat automatically:

1. runs **npm run build** in **../frontend/**,
2. copies **dist/** to **frontend-exe/web/**,
3. creates **.venv** and installs **pywebview** + **pyinstaller**,
4. runs PyInstaller against **SkyVoiceAI.spec**.

Get a Windows `.exe` from a Mac (GitHub Actions)

PyInstaller **cannot cross-compile** — a Windows `.exe` can only be built on Windows, and a macOS `.app` only on macOS. If you're on a Mac and have no Windows machine, let CI build it for you. This folder is self-contained (the built React bundle ships in `web/`), so the workflow needs neither Node nor the private frontend repo.

1. Put this `frontend-exe/` folder in its own GitHub repo and push it:

```
cd frontend-exe
git init && git add . && git commit -m "Sky Voice AI desktop wrapper"
git branch -M main
git remote add origin https://github.com/<you>/sky-voice-ai-desktop.git
git push -u origin main
```

2. The included workflow

`.github/workflows/build-desktop.yml`

runs automatically on every push to `main` (or trigger it from the repo's **Actions** tab → *Run workflow*). It builds on `windows-latest` and `macos-latest` in parallel.

3. Open the finished run and download the artifacts:

- **SkyVoiceAI-windows** → unzip → share the `SkyVoiceAI/` folder; recipients double-click `SkyVoiceAI.exe`.
- **SkyVoiceAI-macos** → unzip → `Sky Voice AI.app`.

Whenever the web app changes, re-run `./build.sh` (or `build.bat`) locally to refresh `web/`, commit it, and push — CI rebuilds fresh binaries.

Configuration

Env var	Purpose
<code>SKYVOICE_URL</code>	Override the URL the window loads. Useful for dev (<code>http://localhost:5173</code>) or to point the desktop app at a staging deployment.
<code>SKYVOICE_DEBUG=1</code>	Open the WebView's devtools (right-click → Inspect on macOS).
<code>SKYVOICE_LOG=DEBUG</code>	Verbose Python logging.

The Cloud Run backend URL itself is **not** configurable here — it's baked into the React bundle at `npm run build` time via Vite's `frontend/.env.production` (`VITE_API_BASE`, `VITE_WS_BASE`). To

re-target the desktop build at a different backend, edit those env vars in `frontend/` and rerun `build.sh`.

App icon

The bundle is already branded with the Sky Voice AI mark — a multi-resolution `assets/icon.icns` (macOS) and `assets/icon.ico` (Windows) are committed, and `SkyVoiceAI.spec` picks the right one per platform automatically. To regenerate them from a new logo:

```
cd frontend-exe
mkdir -p assets

# macOS – convert from the React app's logo:
sips -s format icns ../frontend/public/logo-512.png --out
assets/icon.icns

# Windows – convert from PNG to ICO using ImageMagick or pillow:
python -c "from PIL import Image; Image.open('../frontend/public/logo-512.png').save('assets/icon.ico', sizes=[(256,256),(128,128),(64,64),(32,32),(16,16)])"
```

The spec file (`SkyVoiceAI.spec`) automatically picks `assets/icon.icns` on macOS and `assets/icon.ico` on Windows when they exist.

Permissions

OS	Prompt	First-run behaviour
macOS	Microphone (<code>NSMicrophoneUsageDescription</code>)	System dialog appears the first time the operator clicks Dial . Allowed/denied state persists.
Windows	Edge WebView2 mic	Edge WebView2 surfaces a popup; allow once.
Linux	PulseAudio / PipeWire	Handled by the system mixer; no popup.

Codesigning & distribution (macOS)

Out of the box the .app is **unsigned**, which means Gatekeeper will warn the first time it's launched. For internal use that's fine (right-click → Open). For external distribution:

1. Get a Developer ID Application certificate.
2. Add `codesign_identity="Developer ID Application: Your Name (TEAMID)"` to the `EXE(...)` block in `SkyVoiceAI.spec`.

3. Notarize: `xcrun notarytool submit "dist/Sky Voice AI.app" --keychain-profile=AC_PASSWORD --wait`.

Troubleshooting

- **"missing dependency: pyobjc-framework-WebKit"** on macOS — your system Python is too old. Use Python 3.11+ (Homebrew or python.org).
- **White window** on Windows — install the [Edge WebView2 Runtime](#). Bundled with Win11 by default.
- **Mic permission denied** — quit, then on macOS go to *System Settings* → *Privacy & Security* → *Microphone* and enable Sky Voice AI.
- **App opens but UI is blank** — make sure `web/index.html` exists. Re-run `build.sh`.
- **Voice SDK error 31000 ("getUserMedia")** — the page must be loaded over `127.0.0.1` or `https://`. Don't open `web/index.html` directly from `file://`; let `app.py` start the local server.